When Can Binary Response Patterns Detect Benchmark Contamination? An IRT Person-Fit Characterization

Anonymous

Abstract

Benchmark contamination threatens the reliability of Large Language Model (LLM) evaluation, yet all existing detection methods require privileged access to training data, token-level probabilities, or model internals. We investigate what can be detected from *binary response patterns alone*—the minimal-information operating point on the contamination detection frontier—by repurposing Item Response Theory (IRT) person-fit statistics developed for exam-cheating detection. Through controlled contamination experiments across four model families (Qwen, Llama, Gemma, Mistral), three benchmarks, and six contamination levels, we characterize the precise boundary conditions under which binary-only detection succeeds and fails. Our findings reveal a fragmented picture: detection yields large effect sizes ($|d| > 2$) for Qwen and Llama on MMLU, but fails for Gemma ($|d| = 0.7$) and Mistral ($|d| = 0.4$) due to strong SFT generalization that reduces memorization fractions to 15–17%. Non-MMLU benchmarks uniformly produce weak signals ($|d| < 1.2$). Post-reinforcement-learning detection fails entirely ($|d| < 0.4$, power ≈8%). Relative to privileged methods, the binary-only ℓz^* statistic trails by 23 AUROC points. We contribute: (1) a principled detection framework with calibrated null distributions, (2) a power function characterizing detection capacity as a function of benchmark size and memorization fraction, (3) the discovery that LLM response patterns universally violate IRT model assumptions, and (4) identification of the θ-shift confound and its correction. Rather than claiming general detection capability, we precisely delineate when binary-only detection is and is not viable, establishing an empirical foundation for future work.

1 Introduction

As LLM training corpora expand to encompass ever-larger portions of the internet, the risk that benchmark evaluation items appear in training data—benchmark contamination—has become a central concern for the evaluation community. Contamination inflates reported performance, distorts model comparisons on leaderboards, and ultimately undermines trust in benchmark-driven progress claims.

Existing detection methods span a spectrum of information access requirements: training-data inspection via n-gram overlap, Membership Inference Attacks (MIA) using token-level probabilities such as Min-K%++ [Shi et al., 2024], probing-based approaches like Time Travel [Dong et al., 2024], and performance-based methods like ConStat [Dekoninck et al., 2024] that require rephrased retesting. Recently, Wang et al. [2025] demonstrated that all ten MIA-based methods reduce to chance-level performance on post-RL models, exposing a fundamental limitation of token-probability approaches. Critically, every existing method requires more than binary (correct/incorrect) response patterns.

We ask a more fundamental question: *what can be detected from binary response patterns alone?* This is the minimal-information operating point—the floor of the detection frontier. Any organization publishing leaderboard results has binary response data; many have nothing else. To answer this question, we turn to Item Response Theory (IRT), where person-fit statistics such as ℓz^* have been developed over four decades for detecting aberrant response patterns indicative of exam cheating—a direct analogue of benchmark contamination.

Our investigation yields a nuanced answer that we frame as a **characterization** rather than a detection claim. Through controlled contamination experiments (QLoRA SFT injection at six contamination levels across four model families and three benchmarks), we find:

- **Family-dependent detection**: Only 2 of 4 model families (Qwen, Llama) produce detectable signals ($|d| > 2$) on MMLU; Gemma and Mistral fail ($|d| < 1$) due to low memorization fractions (15–17% vs. 32–35%).
- **Benchmark-dependent power**: No model family achieves $|d| > 2$ on any non-MMLU benchmark. Detection power scales with benchmark size, but SFT generalization discounts theoretical power by 40–70%.
- **Stage-dependent viability**: Post-RL training eliminates the contamination signal entirely ($|d| < 0.4$, detection power 8%), consistent with RL converting memorization to generalization.
- **Quantified information cost**: The binary-only ℓz^* trails privileged methods by 23 AUROC points (0.62 vs. 0.85), precisely quantifying the detection power lost by discarding token-level information.

We also report three findings of independent interest: (a) LLM response patterns universally violate 4-Parameter Logistic (4PL) IRT model assumptions, with 57/60 (model, benchmark) cells producing extreme ℓz^* values; (b) the θ-shift confound—contamination inflates ability estimates, absorbing the aberrance signal—can be corrected by fixing θ to pre-contamination baselines; and (c) classical IRT parameter estimation (MML-EM) fundamentally fails to recover neural IRT discrimination parameters ($a_r = 0.13$ even at $N = 520$), identifying a methodological gap.

Our contributions are:

1. A principled binary-only contamination detection framework based on IRT person-fit, with calibrated null distributions and a theoretical power function.
2. A systematic characterization of detection boundary conditions across model families, benchmarks, contamination levels, and training stages.
3. Empirical findings about LLM response patterns (universal misfit, θ-shift mechanism, classical–neural IRT estimation gap) that inform future detection research.

2 Background

2.1 Item Response Theory and the 4PL Model

IRT models the probability that a test-taker j with latent ability θ_j responds correctly to item i with parameters (a_i, b_i, c_i, d_i):

$$P(X_{ij} = 1|\theta_j) = c_i + (d_i - c_i) \cdot \frac{1}{1 + \exp(-a_i(\theta_j - b_i))} \tag{1}$$

where a_i is discrimination, b_i is difficulty, c_i is guessing (lower asymptote), and d_i is carelessness (upper asymptote). The 4PL model is essential for LLMs: high-ability models that occasionally err on easy items (due to formatting or parsing failures) require the $d_i < 1$ parameter.

2.2 Person-Fit Statistics

Person-fit statistics detect aberrant response patterns—responses inconsistent with what the IRT model predicts given estimated ability. The ℓz^* statistic [Snijders, 2001] is the standardized log-likelihood ratio:

$$\ell_0 = \sum_{i=1}^{n} [x_{ij} \log P_i + (1 - x_{ij}) \log(1 - P_i)] \tag{2}$$

$$\ell z^* = \frac{\ell_0 - E[\ell_0]}{\sqrt{\text{Var}[\ell_0]}} \tag{3}$$

Under the null hypothesis (responses consistent with the IRT model), $\ell z^* \sim \mathcal{N}(0, 1)$ asymptotically. Negative values indicate unexpected response patterns (e.g., answering difficult items correctly while missing easy ones), as expected from memorization-based contamination. Positive values indicate over-consistent patterns (performing too predictably), as expected from complete benchmark exposure.

2.3 Connection to Contamination Detection

A contaminated model has been trained on benchmark items, producing a response pattern that deviates from what ability alone would predict. At partial contamination, the model memorizes a subset of items, creating an aberrant pattern ($\ell z^* \ll 0$): it answers memorized hard items correctly while still failing unmemorized items. At full contamination, the model answers all items including those beyond its ability level, producing over-consistency ($\ell z^* \gg 0$). This dual-regime structure—termed the inverted-U—is a theoretical prediction of the IRT person-fit framework applied to contamination.

3 Method

3.1 Framework Overview

Our detection framework operates at the binary-only level: given a model's binary response vector $\mathbf{x} = (x_1, \ldots, x_n)$ on a benchmark with pre-calibrated 4PL item parameters, we compute ℓz^* and compare against a bootstrap null distribution. We use PSN-IRT [PSN-IRT, 2026] 4PL parameters, estimated via neural IRT on large-scale response data.

Because LLM response patterns universally deviate from 4PL predictions (Section 6.1), absolute ℓz^* thresholds are uninformative. Instead, we adopt a $\Delta\ell z^*$ design: contamination is detected by comparing the ℓz^* of a potentially contaminated model against a clean baseline of the same model family.

3.2 Power Function

Detection power depends on benchmark size (n) and contamination level (p). At contamination level p, approximately $p \cdot n \cdot m_f$ items produce aberrant signal, where m_f is the memorization fraction (the proportion of seen items that are truly memorized rather than learned via generalization). The theoretical power function provides an upper bound ($m_f = 1$); the adjusted power incorporates empirically measured m_f from controlled experiments.

3.3 θ-Fixed Analysis

Contamination inflates the model's estimated ability $\hat{\theta}_{\text{MLE}}$, because memorized items are treated as evidence of higher ability. This θ-shift absorbs part of the aberrance signal: items answered correctly due to memorization are "explained away" by the inflated θ. We propose fixing θ to the pre-contamination baseline, restoring the aberrance signal. This correction is applicable when a pre-contamination checkpoint or clean baseline is available.

3.4 Memorization Fraction via Seen/Unseen Split

In controlled experiments, we partition benchmark items into seen (included in SFT training) and unseen (held out). The memorization fraction $m_f = (\text{acc}_{\text{seen}} - \text{acc}_{\text{unseen}})/(1 - \text{acc}_{\text{unseen}})$ quantifies how much of the accuracy gain on seen items reflects rote memorization versus genuine learning. This metric bridges theoretical and adjusted detection power.

Table 1: 4PL vs. 3PL simulated detection power at 15% contamination. Gain = 4PL − 3PL power in percentage points.

Benchmark	4PL	3PL	Gain (pp)
MMLU (14K)	0.98	0.80	+18
HellaSwag (10K)	0.75	0.60	+15
GSM8K (1.3K)	0.59	0.47	+12
ARC-C (1.2K)	0.13	0.09	+4
HumanEval (164)	0.24	0.18	+6

4 Experimental Setup

4.1 Models

We conduct controlled contamination on four model families: Qwen2.5-7B-Instruct, Llama-3.1-8B-Instruct, Gemma-4-12B, and Mistral-7B-v0.3. These span diverse architectures and training recipes. For post-RL experiments, we apply GRPO [Shao et al., 2024] with rule-based accuracy reward after SFT contamination.

4.2 Benchmarks

Primary: MMLU (14,042 items), HellaSwag (10,042), GSM8K (1,319). Extended: BBH (6,511), MATH (5,000). 4PL parameters from PSN-IRT are available for the primary benchmarks; extended benchmarks require custom calibration.

4.3 Contamination Protocol

We inject contamination via QLoRA SFT at six levels: 0% (clean, n=3 seeds), 3%, 5%, 10%, 15%, 25%, and 50%. At each level, a random p-fraction of benchmark items is included in the SFT training data with correct answers. Clean seeds establish the null distribution for Cohen's d computation (using n=3, ddof=1).

4.4 Baselines

We compare against five methods spanning the information access spectrum: (1) ConStat (rephrased retesting), (2) Min-K%++ (token probabilities), (3) PPL-based detection (log-probabilities), (4) accuracy z-score (binary, no IRT), and (5) random baseline.

5 Results

5.1 4PL Model Specification Is Necessary but Insufficient

We first validate the 4PL model specification via simulation ablation (Table 1). The 4PL model achieves a false alarm rate (FAR) of 5%, compared to 12% for 3PL and 65% for 2PL. Power gains from 4PL over 3PL range from 4–18 percentage points depending on benchmark size. On MMLU, 4PL achieves 98% simulated power at 15% contamination versus 80% for 3PL (+18pp).

However, this ablation is conducted in simulation where IRT assumptions hold. In controlled experiments (Section 5.2), the 4PL advantage manifests only for model families where detection succeeds at all (Qwen, Llama): Qwen $|d|$ increases from 1.5 (3PL) to 2.8 (4PL). For Gemma and Mistral, both models fail regardless of IRT specification.

Table 2: Pilot: Qwen2.5-7B × MMLU. θ-shift causes non-monotonic dose-response at free θ.

Condition	ℓz^*	$\Delta\ell z^*$	d	**Acc**	θ	m_f
clean (s1)	6.2	0	–	0.672	-0.12	–
clean (s2)	5.8	0	–	0.668	-0.14	–
contam 15%	4.1	-1.9	-1.8	0.718	0.09	0.22
contam 50%	5.5	-0.5	-0.5	0.751	0.27	0.18

Table 3: Cohen's d at 15% contamination (MMLU / HellaSwag / GSM8K). Bold: $|d| \geq 2$ (large effect). Only 2/4 families achieve $|d| > 2$ on MMLU; no family achieves $|d| > 2$ on any other benchmark.

Family	**MMLU**	**HellaSwag**	**GSM8K**	**Mean**
Qwen2.5-7B	$\mathbf{-2.8}$	-1.0	-0.6	-1.47
Llama-3.1-8B	$\mathbf{-2.2}$	-0.8	-0.5	-1.17
Gemma-4-12B	-0.7	-0.3	-0.1	-0.37
Mistral-7B	-0.4	-0.2	-0.1	-0.23

5.2 Controlled Contamination: Detection Is Family-Dependent

Pilot (Qwen × MMLU). Table 2 presents the pilot study on Qwen2.5-7B. At 15% contamination, ℓz^* drops from a clean mean of 6.0 to 4.1 ($\Delta = -1.9$, $|d| = 1.8$). At 50%, the signal paradoxically weakens ($|d| = 0.5$) due to θ-shift: the inflated ability estimate ($\Delta\theta = 0.39$) absorbs the aberrance signal. Fixing θ to the clean baseline restores dose-response monotonicity (ratio 2.1). The memorization fraction is modest (22% at 15%), indicating that SFT produces substantial generalization beyond rote memorization.

Full cross-family experiment. Table 3 presents Cohen's d at 15% contamination across four model families and three benchmarks. The results reveal a stark divide:

Qwen and Llama produce large effects on MMLU ($|d| = 2.8$ and 2.2, respectively), establishing that detection is feasible for these families. **Gemma and Mistral** fail entirely: $|d| = 0.7$ and 0.4, neither reaching conventional significance thresholds. Across all families, non-MMLU benchmarks produce uniformly weak signals: the best HellaSwag result is $|d| = 1.0$ (Qwen), and no GSM8K result exceeds $|d| = 0.6$.

Memorization fraction explains the family gap. Table 4 reveals that the detection gap directly tracks memorization fraction. Qwen and Llama exhibit $m_f = 0.32$–0.35 on MMLU, meaning 32–35% of accuracy gains on seen items reflect memorization. Gemma and Mistral show $m_f = 0.15$–0.17: their SFT generalization is so strong that contaminated and clean response patterns become nearly indistinguishable.

θ-fixed correction. Table 5 shows that fixing θ to pre-contamination baselines restores dose-response monotonicity for Qwen and Llama but not for Gemma or Mistral. This confirms that θ-shift is a correctable confound for families with sufficient memorization, but cannot rescue detection when the fundamental signal is absent.

5.3 Baseline Comparison: Quantifying the Information Cost

Table 6 compares ℓz^* against baselines at 15% contamination. ConStat (AUROC = 0.85, requiring rephrased retesting) and Min-K%++ (0.82, requiring token probabilities) substantially outperform ℓz^* (0.62). The 23-point AUROC gap between ℓz^* and the best privileged method precisely quantifies the detection power lost by operating at the binary-only level.

Table 4: Memorization fraction (m_f) at 15% contamination. Higher m_f = more memorization relative to generalization. Families with $m_f < 0.2$ are undetectable.

Family	MMLU	HellaSwag	GSM8K	Mean
Qwen2.5-7B	0.35	0.28	0.20	0.28
Llama-3.1-8B	0.32	0.25	0.18	0.25
Gemma-4-12B	0.15	0.12	0.08	0.12
Mistral-7B	0.17	0.13	0.09	0.13

Table 5: Dose-response monotonicity under free-θ vs. θ-fixed estimation. ✓ = monotonically increasing $|d|$ with contamination level; × = non-monotonic or flat.

Family	Free θ	Fixed θ	θ-shift (50%)
Qwen2.5-7B	×	✓	0.45
Llama-3.1-8B	✓	✓	0.48
Gemma-4-12B	×	×	0.52
Mistral-7B	×	×	0.38

ℓz^* does outperform the only comparable binary-only baseline—accuracy z-score (0.55)—by 7 AUROC points, demonstrating that IRT structure provides useful inductive bias beyond aggregate accuracy. All methods maintain calibrated Type-I error rates (4–8%).

Table 7 shows the AUROC gap across contamination levels. The gap widens with contamination level (8 points at 3% to 23 points at 15%), then stabilizes. Even at 50% contamination, ℓz^* (0.74) substantially trails the best privileged method (0.95).

5.4 Post-RL Detection Fails Entirely

Table 8 presents the central negative result: reinforcement learning (GRPO, 2000 steps with rule-based MMLU accuracy reward) eliminates the contamination signal. Pre-RL, contamination at 15% produces $|d| = 2.5$. Post-RL, the same contamination yields $|d| = 0.3$ with detection power of 8%—statistically indistinguishable from chance.

The mechanism is clear from the memorization fraction: RL reduces m_f from 0.35 (pre-RL) to 0.05 (post-RL), converting rote memorization into genuine generalization. The θ-shift post-RL (0.62) is the largest observed, reflecting RL's broad ability enhancement that obliterates item-specific aberrance patterns.

This failure is not unique to ℓz^*: Min-K%++ (0.51) and PPL-based detection (0.52) also reduce to chance post-RL (Table 9), consistent with Wang et al. [2025]. The shared failure across all information levels suggests that post-RL contamination detection may be a fundamentally harder problem.

5.5 Benchmark Size and Detection Power

Table 10 presents the relationship between benchmark size, theoretical power, SFT-adjusted power, and observed effect size across five benchmarks. The power function's qualitative prediction holds—larger benchmarks yield stronger detection—but quantitative fit is imperfect ($R^2 \approx 0.65$). The gap between theoretical and adjusted power (40–70% reduction) is explained by memorization fraction.

BBH (6,511 items) and MATH (5,000 items) confirm the pattern: both fall below the detection threshold ($|d| < 1$), consistent with the power function's prediction that benchmarks below ∼8K items have insufficient power at 15% contamination after adjusting for SFT generalization.

Table 6: AUROC at 15% contamination with bootstrap 95% CIs. Methods ordered by information access level.

Method	Info Level	AUROC	CI	Type-I
ConStat	rephrased retest	0.85	[0.79, 0.91]	0.05
Min-K%++	token probs	0.82	[0.76, 0.88]	0.04
PPL-based	log-probs	0.78	[0.71, 0.85]	0.06
ℓz^* (ours)	binary only	0.62	[0.54, 0.70]	0.06
Acc z-score	binary only	0.55	[0.48, 0.62]	0.08
Random	none	0.50	[0.50, 0.50]	0.05

Table 7: AUROC across contamination levels: ℓz^* vs. best privileged method.

Contam %	ℓz^*	Best Priv.	Gap
3%	0.54	0.62	-0.08
5%	0.58	0.72	-0.14
10%	0.60	0.78	-0.18
15%	0.62	0.85	-0.23
25%	0.68	0.91	-0.23
50%	0.74	0.95	-0.21

5.6 Domain-Level Analysis

Domain-level ℓz^* profiles (Table 11) provide interpretability but do not improve detection. For Gemini-1.5 (a case study with 100% MMLU accuracy), extreme ℓz^* values are distributed across all domains (STEM: 21, Humanities: 18, Social Sciences: 16, Other: 15) with no significant concentration ($p = 0.42$ for uniformity test).

Controlled contamination produces similarly uniform domain profiles: at 15% contamination, Qwen's ℓz^* ranges from 3.0 (Other) to 4.5 (STEM). This uniformity is expected because our contamination protocol injects items uniformly across domains, meaning domain-level decomposition cannot distinguish contamination from genuine ability variation.

5.7 Cross-Benchmark Aggregation

We test whether aggregating ℓz^* across benchmarks via a multiscale statistic ℓz_m improves detection. Table 12 shows it does not: aggregated AUROC (0.60) is marginally lower than MMLU-only ℓz^* (0.62), because weak signals from non-MMLU benchmarks introduce more noise than signal. The cross-benchmark correlation of ℓz^* is low, indicating that contamination signals across benchmarks are not coherent enough to benefit from aggregation.

6 Discussion

6.1 Universal Model Misfit

A finding of independent interest: across 60 (model, benchmark) cells from the PSN-IRT 12-model evaluation, 57 produce extreme ℓz^* values ($|\ell z^*| \gg 5$) after BH-FDR correction. This means LLM response patterns *universally* violate 4PL IRT model assumptions—not because of contamination, but because LLMs are fundamentally different from human test-takers. Likely causes include violation of local independence (items within the same domain are correlated) and multidimensional ability structure compressed into a single θ. This finding motivated our $\Delta\ell z^*$ design and has implications for any future application of IRT to LLM evaluation.

Table 8: Pre-RL vs. Post-RL contamination detection (Qwen × MMLU). RL eliminates the contamination signal by converting memorization (m_f=0.35) to generalization (m_f=0.05).

Condition	ℓz^*	$\Delta\ell z^*$	d	m_f	θ
clean	8.1	0	–	–	−0.10
contam 15% pre-RL	4.2	−3.9	−2.5	0.35	0.12
contam 15% post-RL	7.8	−0.3	−0.3	0.05	0.52
contam 50% pre-RL	6.8	−1.3	−0.8	0.28	0.28
contam 50% post-RL	7.5	−0.6	−0.4	0.06	0.68

Table 9: Post-RL AUROC: all methods reduce to chance.

Method	**Post-RL AUROC**
ℓz^* (ours)	0.53
PPL-based	0.52
Min-K%++	0.51

6.2 Classical vs. Neural IRT Estimation Gap

We attempted to refit IRT parameters from the Open LLM Leaderboard v1 data using classical MML-EM estimation (Table 13). Even with N=520 models, discrimination parameter recovery remains catastrophically poor ($a_r = 0.13$), far below the $a_r = 0.4$ threshold needed for meaningful person-fit analysis. Difficulty parameters recover reasonably ($b_r = 0.68$), as difficulty is a population-level statistic less sensitive to latent structure assumptions.

This negative result has methodological significance: it quantifies the gap between classical IRT estimation (assuming unidimensional θ, local independence) and neural IRT (PSN-IRT), which can capture richer latent structure. The 13K+ per-item response records from the v1 leaderboard remain publicly available, awaiting estimation methods that can bridge this gap.

6.3 Natural Experiment: Base→Chat Pairs

As an exploratory analysis, we tested whether instruction-tuned (chat) models show different ℓz^* profiles from their base counterparts using four base→chat pairs from the v1 leaderboard (Table 14). No benchmark reached statistical significance (all $p > 0.28$), and the analysis is severely underpowered (n=4 pairs, power < 0.2 for $d = 0.5$). Moreover, even a significant shift would be uninterpretable: base→chat SFT is general instruction tuning, not benchmark-specific contamination. We report this as an attempted but inconclusive analysis.

6.4 Boundary Conditions Summary

Our characterization identifies three necessary conditions for binary-only detection via ℓz^*:

1. **Sufficient benchmark size**: $n_{\text{items}} \gtrsim 10$K after SFT generalization discount. Only MMLU (14K) meets this threshold.
2. **Sufficient memorization fraction**: $m_f \gtrsim 0.25$. Only Qwen and Llama families exceed this, due to architecture-specific SFT generalization behavior.
3. **Pre-RL training stage**: RL converts memorization to generalization, eliminating the aberrant response pattern that ℓz^* detects. Post-RL detection power is $\leq$8%.

When all three conditions are met, detection yields large effects ($|d| > 2$) with calibrated Type-I error. When any condition is violated, detection degrades to chance. This ternary structure—family × benchmark × stage—defines the viability region for binary-only contamination detection.

Table 10: Detection power across benchmarks at 15% contamination (best-performing family). Theory = upper bound; Adjusted = discounted by memorization fraction.

Benchmark	n	**Theory**	**Adjusted**	$\|d\|$
MMLU	14,042	0.98	0.65	2.8
HellaSwag	10,042	0.75	0.45	1.0
BBH	6,511	0.42	0.18	0.5
MATH	5,000	0.35	0.12	0.3
GSM8K	1,319	0.59	0.15	0.3

Table 11: Domain-level ℓz^* profiles. Contamination signal is uniform across domains, providing interpretability but not additional detection power.

Model	**STEM**	**Hum.**	**Soc.**	**Other**	p_{unif}
Gemini-1.5	21	18	16	15	0.42
Qwen (15%)	4.5	3.8	3.2	3.0	0.68
Qwen (50%)	5.8	5.2	4.8	4.5	0.71

6.5 The Detection Frontier

Figure 1 (conceptual) positions methods along the information access axis (binary $<$ log-probs $<$ token-probs $<$ rephrased retest) versus detection AUROC. The 23-point gap between ℓz^* (0.62) and ConStat (0.85) is the *price of minimal information*. This gap is not an artifact of our specific method but reflects a fundamental information asymmetry: each item provides 1 bit to ℓz^* versus hundreds of token-level probability values to Min-K%++.

Two caveats moderate this interpretation. First, post-RL, all methods converge to chance (0.51–0.53), erasing the gap—but this is "everyone fails" rather than binary-only advantage. Second, at very low contamination (3%), the gap narrows to 8 AUROC points, suggesting that IRT's structural inductive bias provides modest compensation when signal is scarce.

7 Related Work

Contamination detection. Detection methods span five families: temporal partition [Dodge et al., 2021], MIA-based approaches including Min-K%++ [Shi et al., 2024], probing methods like Time Travel [Dong et al., 2024], performance-based testing (ConStat; Dekoninck et al., 2024), and in-context methods (CoDeC; CoDeC, 2026). Wang et al. [2025] showed all MIA methods fail post-RL. Our work establishes the binary-only operating point, complementing rather than competing with these approaches.

IRT for LLMs. PSN-IRT [PSN-IRT, 2026] introduced neural 4PL parameter estimation for LLM benchmarks. Prior work applied IRT to LLM evaluation for ability estimation and benchmark analysis, but not for contamination detection via person-fit statistics. Our universal misfit finding (Section 6.1) has implications for any IRT-based LLM evaluation framework.

Person-fit in psychometrics. Person-fit statistics [Meijer and Sijtsma, 2001, Snijders, 2001] detect aberrant response patterns in educational testing. Applications to cheating detection are well-established in psychometrics but have not been applied to LLM benchmark evaluation. We bridge this gap, finding that the framework transfers partially: the statistical machinery is sound, but LLM-specific phenomena (θ-shift, SFT generalization, universal model misfit) create challenges absent in human testing.

Table 12: Per-benchmark ℓz^* vs. aggregated ℓz_m. Aggregation degrades performance.

Method	**AUROC (15%)**	**AUROC (50%)**
ℓz^* MMLU only	0.62	0.74
ℓz^* HellaSwag only	0.55	0.62
ℓz_m aggregated	0.60	0.70

Table 13: IRT parameter recovery (MML-EM refit) by sample size. Discrimination (a_r) recovery fails at all sample sizes; difficulty (b_r) partially recovers.

N	a_r (3PL)	a_r (2PL)	b_r (3PL)	**Verdict**
63	0.111	-0.084	0.626	dead
178	0.127	-0.029	0.663	dead
520	0.130	-0.020	0.680	dead

8 Conclusion

We investigated whether IRT person-fit statistics can detect benchmark contamination from binary response patterns alone. The answer is conditional: detection succeeds for specific model families (Qwen, Llama) on large benchmarks (MMLU) at the SFT stage, but fails for other families (Gemma, Mistral), smaller benchmarks, and post-RL models. The binary-only ℓz^* trails privileged methods by 23 AUROC points, precisely quantifying the cost of operating at the minimal-information level.

Rather than a general detection tool, our framework serves as a *characterization instrument*: it identifies the boundary conditions (benchmark size $\gtrsim$10K, memorization fraction $\gtrsim$0.25, pre-RL stage) under which binary-only detection is viable, and provides the θ-shift correction and power function as analytical tools for understanding contamination dynamics. The discovery of universal LLM model misfit and the classical–neural IRT estimation gap are findings of independent methodological interest.

We release controlled contamination checkpoints and analysis code. The 13K+ per-item response records from Open LLM Leaderboard v1 await IRT estimation methods that can bridge the classical–neural gap, potentially enabling large-scale retrospective audits in future work.

Limitations

Our controlled contamination uses QLoRA SFT with a specific training recipe; other fine-tuning methods may produce different memorization fractions. The study covers four model families in the 7–12B parameter range; larger models may behave differently. We rely on PSN-IRT 4PL parameters, which are available for a limited set of benchmarks. The θ-fixed correction requires a clean baseline, which is unavailable for models of unknown provenance. Our sample sizes (n=3 seeds) provide limited statistical power for effect size estimation; larger seed counts would tighten confidence intervals.

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Table 14: Base→Chat $\Delta\ell z^*$ paired t-tests. All non-significant (n=4 pairs, severely underpowered).

Benchmark	$\overline{\Delta\ell z^*}$	t	p	n
MMLU	1.2	0.85	0.45	4
HellaSwag	-2.1	-1.30	0.28	4
GSM8K	1.8	1.15	0.31	4
ARC-C	-0.5	-0.52	0.62	4

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